General Science

General Science: Grade 6

GRADE(S): 6

About the Course

Learners are introduced to scientific activities and technology with a nod toward historical context. Specifically, students learn about plate tectonics, anatomy and medicine, and architectural engineering. The lab component is required for this course. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 6

About Science: Grade 6
In level 6, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from oceanography, architecture, and more. Reading level and time required are transitional as we gently prepare for the middle grades.

Science: Grade 6
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Building on the familiarity and foundational knowledge of Form 1, Form 2 learners in grades 4-6 extend the scope of the Things they encounter and begin to notice that science is active in their world. Growing in their self-knowledge at this age, they may notice that they gravitate toward some of these Things more than others. This is a bit like noticing that you have a natural rapport with a particular friend because they seem to understand the way you think and share the same interests. This is wonderful and very typical, but we still want them to get to know more “friends” in science and try different Things, even though one might be their particular interest. Their sense of community is expanding, as well: who are the scientists in their world, and what are they doing? Special care has been taken to build vocabulary and conceptual knowledge “by the way” and to maintain the connection between scientific endeavors and the human experience. In Form 3, students will continue to build on this knowledge of what science is doing in the world when they examine science in its historical context.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

About Science: Grade 6
In level 6, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from oceanography, architecture, and more. Reading level, historical context, and time required are transitional as we gently prepare for the middle grades.

General Science: Grade 6
For sixth-grade students or possibly hungry fifth-graders or seventh-graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, or teachers can adapt the reading level and laboratory content to meet their needs.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
6	General Science: Grade 6 2 times/week 30 min	Plate Tectonics The Art of Construction Phineas Gage

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 6				
Natural History: Grade 6		General Science: Grade 6 Nature Walks & Scouting: Grades 1-8	General Science: Grade 6 Nature Notebook: Grade 6	Labs: Grade 6

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 6
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

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Encourage students to begin choosing their own special topic to observe, study, and research in their field guides. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too! Suggested field trips include:

General Science: Grade 6
☐ Term 1: any local geology site (e.g. a quarry or certain parks)
☐ Term 2: hospital open house
☐ Term 3: any buildings or bridges

Term Prep & Teacher Tips

Science: Grade 6
Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term and/or any that support special topics chosen by students:

General Science: Grade 6
☐ Term 1: specific rock types p.748-759
☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
• less than 1/2 c white vinegar
• recycled cardboard, roughly 3' x 3'
• books or wood blocks to adjust height

- 3 colored markers
- 2 or more apples (Term 2)
- plate
- knife
- cutting board
- plastic wrap
- roughly 2' x 2' piece of substrate (recycled cardboard, wood scrap, foam core)
- 15 sheets of multipurpose paper
- any weights for testing bridge
- various supplies by student design
- printables
- computer
- optional: camera, such as found on most electronic devices
- optional: antiseptic treatments
- optional: plastic ruler
- optional: fabric scrap
- optional: wax paper
- optional: fan/hair dryer
- optional: needle and thread
- optional: cereal box to model a building
- optional: rocks
- optional: sand paper
- optional: 3-7 weights with a hole, e.g. washers or spools
- optional: string or strong thread
- optional: hammer and nail
- optional: block of scrap wood
- optional: rope
- optional: nutcracker
- optional: 4 raw eggs (Term 3)
- optional: 8 sturdy egg cups
- optional: 8 cotton balls
- optional: scrap plywood (might be the same substrate as above)
- optional: cylindrical balloon
- optional: cup that fits the top of balloon
- optional: books of various sizes

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details]

General Science: Grade 6

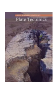
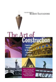
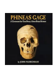


Plate Tectonics



The Art of Construction



Phineas Gage

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ View Basic Supplies

∞ View Supply List Details

General Science: Grade 6

Alveary Grade 6 Science Kit

Household Items - General Science: Grade 6

 Student Microscope

 Prepared Microscope Slides

Quick Links

Related Course Quick Links

- Extra Helpings
- Outdoor Work
- Seek app from iNaturalist
- SkyView Lite for iOS
- SkyView Lite for Android
- Foundations (See Section 13: Science)

Click THIS text or scan the QR code for links.

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How To Teach

Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there were any dates that they want to keep for their Book of Centuries.

Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

- These can be viewed as alternative ways to engage.

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Click THIS text or scan the QR code for links.

Term 1

WEEK 1 ⬚ **30m General Science: Grade 6 - Lesson 1**

God Created

▯ Materials: The Holy Bible

➔ INTRO

You have been learning about scientific activities in our modern world. Like all knowledge, our culture affects these activities, and in return, what we learn affects our culture. This has been going on throughout human history. Early scientists primarily studied the Things they could see and assumed that they were always this way, but eventually, they began to make some curious observations that suggested that maybe Things hadn't always been this way. These observations were shocking and confusing to these men and women. This was a major discovery that led many naturalists to question all they thought they knew and understood. It led some to reject science; others to reject their faith in God; and still others to be awestruck at the elegance and majesty of the Creator. What were these discoveries all about, and how did they change the direction of science? These are some very big ideas that require a lot of time to think about. Before getting started with the science, let's spend some time with our Bibles.

➔ READ, NARRATE, & DISCUSS

Genesis 1-2

• What is the same in Genesis 1 and 2, and what is different? How do these two chapters work together to tell us about God?

★ TEACHER NOTE

This lesson consists of reflection on the Christian story of creation. Teachers might choose to allow students to do/review this independently, to engage in the reading/discussion, or to skip this one.

WEEK 1 ⬚ **30m General Science: Grade 6 - Lesson 2**

Making Sense of What We See

▯ Materials: Plate Tectonics

➔ INTRO

Some of the discoveries that made scientists think that maybe Things changed over time came from studying rocks. Look at the title and cover. Do you have ideas about plate tectonics? We'll begin the story today with some of those early scientists.

Notice the sidebars in the text, like the one on p.14-15. As you become more independent with your reading, it can be helpful to think about how to deal with these kinds of "rabbit trails," which can enhance the reading. Because it interrupts the main line of thought, it is usually best to finish the section and then come back to the sidebar.

➔ READ & NARRATE

Ch.1 p.11-13, 16-17 "From some" - "to be connected."

➔ READ, NARRATE, & DISCUSS

p.14-15 "Around the" - "region's geology."

WEEK 2 ⬚ **30m General Science: Grade 6 - Lesson 3**

Seismic Disasters

▯ Materials: Plate Tectonics

➔ RECAP

Recall what we read in the last passage. Today we'll read about seismic activity in Italy and Japan.

➔ READ & NARRATE

∞ Article Link: The Destruction of Pompeii

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[Click THIS text or scan the QR code for links.]

Term 1

→ READ, NARRATE, & DISCUSS
p.17-20 “Pompeii” - “with the natural world.”

WEEK 2 **30m General Science: Grade 6 - Lesson 4**
Renaissance Ideas

Materials: Plate Tectonics

PREP: Read Teacher Tip

→ RECAP
Tell me about the occurrences in Italy and Japan. Naturalists continued to observe for a long time, but then our story leaps forward during the Renaissance.

→ READ, NARRATE, & DISCUSS
p.20-27 "After the decline" - "centuries earlier."

• Nicolaus Steno's work presented, for the first time, the idea that the Earth and even life itself were dynamic. Have you ever learned something so new and surprising that it seemed to change everything? What did you do with this new information?
• While Steno did abandon his scientific pursuits when he entered religious life, many others then and now pursue scientific endeavors and religious vocations hand-in-hand. Daniello Bartoli, Paolo Casati, and Maria Gaetana Agnesi are just a few from Steno's time period. How many can you find?

→ SUPPLEMENTAL
The author notes that ideas often inspire people who come later. One scientist you should know who is not mentioned in our book is Mary Anning:
∞ Video Link: Mary Anning (3:40)

★ TEACHER NOTE
Conflict between religion and Steno's work is alluded to on p.25: "Like many of" - "remainder of his life." Teachers might choose to allow it to pass by the way for now (this idea is a major theme in Grade 7-8), to engage in discussion, or to check out the book about Steno in Extra Helpings. The same conflict is mentioned in the supplemental video (1:16-1:31).

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 3 **30m General Science: Grade 6 - Lesson 5**
Rocks and Hutton's Theory

Materials: Plate Tectonics

→ RECAP
Tell me about what Renaissance thinkers brought to the story. Before we read about the theory of plate tectonics, look at the figure on p.28. You may continue to look back at it during the passage.

→ READ, NARRATE, & DISCUSS
p.29-35 "We now know" - "parts of the puzzle?"

• Discuss the main points of James Hutton's theory. Use words, pictures, or diagrams.

★ TEACHER NOTE
The age of the Earth is mentioned along with Hutton's Theory of the Earth on p.33-34: "He also claimed" - "wind or rain." Teachers might engage in discussion, present alternative theories, or check one of the books in Extra Helpings for additional discussion.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 3 **30m General Science: Grade 6 - Lesson 6**
Oceanography and Seismology

Materials: Plate Tectonics

→ RECAP
Recall the last reading. We've read about paleontology and geology. Today we have two more types of study that will reveal more puzzle pieces: oceanography and seismology.

→ READ & NARRATE
p.35-36 "Exploring the Oceans" - "sea level."

Term 1

→ **READ, NARRATE, & DISCUSS**
p.36-41 "Finding ways" - "accurate measurement."

